

Mahasamvad AI Newsroom

Production Architecture & Operational Platform

A comprehensive technical and operational overview of the sovereign, Marathi-first generative AI content platform engineered and deployed for the Directorate General of Information & Public Relations (DGIPR), Government of Maharashtra.

1. Executive Summary & Strategic Rationale

The Directorate General of Information and Public Relations (DGIPR) is the primary communications bridge between the Government of Maharashtra and more than 12 crore citizens. The department is mandated to convert high-stakes Cabinet decisions, Government Resolutions (GRs), ministerial press briefings, Doordarshan studio interviews, and district field updates into accurate public communication across multiple formats and languages. This work must often be completed within hours of an official meeting, while maintaining uncompromising factual accuracy, nuanced administrative Marathi, correct official designations, and state visual identity.

Historically, this workflow faced severe operational bottlenecks. Official information reaches the department in fragmented, unstructured formats: handwritten notes, WhatsApp voice memos, multi-page scanned PDF resolutions, photographed letters, and live broadcast footage. Departmental teams spent 4 to 6 hours manually listening, transcribing, cross-checking ministerial portfolios, drafting news summaries, designing social creatives, and producing explainer videos. This manual pipeline created unavoidable delays, factual drift, inconsistent Devanagari spelling, and template mismatches during urgent announcements. Furthermore, feeding sensitive or unpublished government material into public cloud AI tools posed serious data sovereignty, confidentiality, and jurisdiction risks.

To solve these challenges, **HashCase (HashByte Pvt Ltd)** engineered and deployed the **Mahasamvad AI Newsroom**—a sovereign, private generative AI ecosystem operating entirely within government-controlled infrastructure. The platform treats official primary documents as the immutable source of truth, using Retrieval-Augmented Generation (RAG), verified terminology glossaries, and deterministic code-side formatting engines to synthesize publication-ready Marathi articles, social posters, YouTube thumbnails, and branded explainer videos in minutes under strict human-in-the-loop oversight.

Core Mandate & Strategic Value

By unifying source intake, audio transcription, factual drafting, visual design, and video production into a single governed workflow, the platform compresses publication turnaround from over 4 hours to under 15 minutes, while guaranteeing a 100% verifiable lineage back to original government documents.

2. Sovereign Architecture & Technical Foundation

The AI Newsroom is engineered from the ground up as an enterprise-grade, data-sovereign system. Rather than treating AI models as unconstrained text generators, the platform enforces strict boundary controls across four fundamental architectural layers:

INFRASTRUCTURE

Air-Gapped & On-Premises Privacy

All language, vision, speech, and embedding models operate within infrastructure controlled by DGIPR or approved state data centers. State documents, audio recordings, and vector embeddings never leave authorized government infrastructure, fully complying with the Digital Personal Data Protection (DPDP) Act 2023.

KNOWLEDGE BASE

Historical Archive & Editorial Style Matching

Historical Mahasamvad articles are indexed in a private database that understands context and structure. The retrieval system finds past published articles to guide the writing tone, layout, and editorial style of new drafts, while ensuring that old facts, dates, or figures are never mixed into current government releases.

LINGUISTICS

Two-Tier Verified Glossary

A central administrative glossary maintains verified Marathi, Hindi, and English spellings for ministers, departments, and government schemes. Automated entity resolvers map spoken colloquial audio transcripts (such as *हायर अँड टेक्निकल* to *उच्च व तंत्रशिक्षण मंत्री*) directly to official portfolios.

DETERMINISTIC

Code-Side Guardrail Enforcement

Formatting constraints, scene durations, and schema contracts are enforced strictly in code via structured output schemas, image composition libraries, and video stream processors, ensuring predictable, production-ready outputs rather than relying on natural-language model prompts.

3. Multimodal Ingestion & Mobile Integration

A major engineering achievement of the platform is eliminating manual data transfers between mobile devices and desktop workstations. The web application is built as an installable Progressive Web Application (PWA) with native system hooks:

- Direct Android Share Sheet Integration:** The application registers a native audio share target on supported Android devices. When an officer receives a WhatsApp voice note or audio recording, they can share it directly into the platform with a single tap, bypassing the need to download or manually locate files in phone storage.
- Client-Side Service Worker Caching:** A specialized background service worker intercepts incoming audio files on the phone, buffers recordings in secure browser storage, and opens the transcription workspace immediately without routing uncompressed audio through intermediate third-party servers.
- Universal Audio Format Support:** Direct processing across all common formats, including WhatsApp voice notes (OPUS), MP3, M4A, AAC, WAV, FLAC, and WebM, matching enterprise speech recognition standards.
- Controlled Page-Range Document Selection:** For long government resolutions and multi-page PDFs, officers can inspect page thumbnails and select only the relevant pages before optical character recognition (OCR) begins. This eliminates unnecessary processing latency on blank pages, cover sheets, or administrative annexures.

4. Core Production Modules Delivered & In Active Use

The platform translates disparate government inputs into six specialized production workspaces, each equipped with dedicated review cards, revision controls, and channel-ready export formats:

MODULE / WORKSPACE	INPUT MEDIA ACCEPTED	OPERATIONAL CAPABILITIES & DELIVERABLES	PRODUCTION TURNAROUND
1. Meeting & Document-to-Article DLO / लेख निर्मिती	Cabinet notes, multi-page scanned GRs, typed briefs, photographed letters.	Extracts 5W1H factual summaries (Who, What, When, Where, Why, How), identifies officials and portfolios, matches Mahasamvad editorial style, and produces concise news reports or detailed explanatory articles.	First Draft: < 15 mins Revision: < 2 mins Factual Accuracy: 100%
2. Audio & YouTube Transcription ध्वनिलेखन	WhatsApp voice notes (OPUS), M4A, MP3, AAC, and direct YouTube URLs.	Transcribes recordings and online press briefings concurrently, segments speech with timestamps, provides speaker attribution, and sends text directly into the drafting workflow without re-uploading media.	1-Hour Audio: < 4 mins YouTube Stream: Direct Accuracy: Domain-tuned
3. Governed Social Creatives क्रिएटिव्ह व पोस्टर्स	Approved Marathi articles, policy announcements, scheme targets.	Selects approved design templates based on text volume, applies official color palettes, creates matching captions, and exports three-layer Canva presentation packages with protected government branding.	Initial Graphic: < 45 secs Element Redraw: < 20 secs Format: Multi-aspect
4. Branded Explainer Videos व्हिडिओ निर्मिती	Factual notes, approved Marathi scripts, audio voiceovers.	Generates multi-scene video storyboards with each scene strictly capped at 5 seconds, supports custom officer reference photos, enables single-scene redraws, and appends the official closing contact slate.	Storyboard: < 2 mins Full Video: < 8 mins Scene Redraw: < 45 secs
5. Multilingual Translation & Proofing भाषांतर व मुद्रितशोधन	Marathi press notes, government circulars, PDF documents.	Translates Marathi into Hindi and English while preserving proper names, designations, and scheme titles; proofreads Marathi text to highlight orthographic spelling and grammar errors with side-by-side comparisons.	1,000 Words: < 60 secs Proper Nouns: Preserved Numerical Accuracy: 100%
6. Native PDF Chat & Work Assistant संभाषण सहाय्यक	Original PDF files (up to 50MB), images, text queries.	Provides an interactive workspace for summarizing forwarded GRs, extracting complex tables, answering queries across long reports, and drafting official correspondence with persistent multi-turn context.	PDF Ingestion: Direct Context: Multi-turn Model: Enterprise Quality

Connected Work Threads & Content Reuse

Completed assignments form connected operational threads. An approved article can instantly be turned into a social poster, caption, or video script without re-uploading or re-processing the original source documents.

5. Key Engineering Safeguards & Design Decisions

To ensure generative AI remains reliable, compliant, and visually consistent for official government publication, HashCase developed four specialized technical safeguards:

5.1 Matching Content Volume to Design Layouts

When generating visual social creatives, automated systems often make the mistake of squeezing too much text into a small graphic, or stretching a short two-line announcement across a complex template with empty placeholders.

In this platform, the system first analyzes the input text and counts how many distinct points or sentences need to be shown. It then matches that content against pre-approved design templates in the library, choosing a layout designed specifically for that amount of text. If a note contains four points, it selects a four-point layout rather than a template built for eight points or two points. This prevents text from being shrunk to unreadable sizes, cut off mid-sentence, or artificially padded with empty space.

5.2 Protecting Official State Branding & Layered Canva Export

AI image models cannot be trusted to draw official government emblems or department logos—they frequently distort the emblem, mistype lettering, or alter state colors. To prevent this, the platform handles branding purely through code:

- **Dedicated Keep-Out Areas:** Generated artwork is instructed to leave clean, unobstructed areas at the top-right (for the state emblem) and bottom (for the department footer), rendered as natural continuations of the background.
- **Three-Layer Canva Export:** When exporting to Canva for further design adjustments, the system separates the final graphic into three distinct layers inside a presentation file: the top emblem, the bottom footer, and the editable artwork in between. Design teams can edit or adjust the artwork in Canva without any risk of altering, moving, or distorting official state branding.

5.3 Five-Second Scene Narration & Video Pacing

Rather than asking an AI model to guess how long a video should be, the video pipeline follows a strict pacing rule: every scene is limited to a maximum of five seconds of spoken Marathi narration.

The narration text is analyzed and split into individual scene lengths based on natural speech pacing. This guarantees that each visual scene transitions smoothly with the spoken voiceover, avoiding rushed audio, awkward pauses, or desynchronized footage. Every video is finalized with synchronized subtitles, a crisp department wordmark, and the official 2.18-second DGIPR contact slate.

5.4 Using Real Reference Photographs for Video Scenes

Government announcements often center on a specific real-world subject—such as a newly constructed bridge, a hospital building, an irrigation project, or an official logo. Describing these in text alone can lead an image model to generate a generic or inaccurate depiction.

The video workspace allows officers to attach a real photograph directly to any scene. The system pre-processes the photo (standardizing format, orientation, and resolution) and uses it to guide the visual model. The resulting video frame accurately reflects the actual building or location while generating natural cinematic composition and lighting appropriate for the scene.

6. Department-Level Telemetry & Utilization Tracking

To provide leadership with transparent operational visibility without monitoring individual officers, the platform includes a native telemetry engine that records processing activity in a central audit log:

USER-FACING TASK	UNDERLYING TECHNOLOGY	PROCESSED UNIT RECORDED	OPERATIONAL ANALYTICS PURPOSE
Audio & YouTube Transcription	Speech-to-Text Engine	Exact audio duration (minutes & seconds)	Tracks transcription volume across press briefings and meetings.
Scanned Document Reading	Vision OCR Pipeline	Processed page count	Monitors volume of scanned GRs and circulars ingested.
Article Drafting	Language Model Stack	Input / Output token count	Measures drafting output across news releases and feature articles.
Social Creatives & Posters	Visual Layout Engine	Rendered graphics count	Audits design output across social media channels.
Explainer Video Production	Video & Speech Synthesis	Scene count & narration seconds	Tracks storyboard creation, clip renders, and scene redraws.
Translation & Proofreading	Multilingual Stack	Word count and token volume	Measures translation throughput and proofing activity.

7. Operational Governance & On-Site Enablement

Successful adoption of generative AI within government administration requires robust operational governance and hands-on enablement:

- Dedicated On-Site Operational Support:** HashCase deploys trained professionals on-site at DGIPR headquarters to provide continuous daily assistance, handle urgent Cabinet releases, and train departmental staff directly.
- Immutable Version History:** Every article draft, social graphic revision, and video iteration is saved as an immutable version. Officers can review changes and revert to any prior state instantly, ensuring creative experimentation never jeopardizes approved work.
- Review-Driven Publishing Handoff:** Nothing moves automatically to public release. The platform presents structured review cards at every stage, leaving final publication and distribution authority strictly with authorized departmental officers.
- Complete Data Sovereignty:** All generated articles, social assets, video files, transcripts, and custom glossaries remain under the exclusive ownership and control of DGIPR, stored on authorized state infrastructure.

Summary & Long-Term Institutional Capability

The Mahasamvad AI Newsroom establishes a sovereign, production-hardened AI content operations system built specifically for the day-to-day requirements of government communication in Maharashtra. By combining strict factual grounding, automated layout intelligence, and continuous on-site operational support, DGIPR now operates an end-to-end newsroom capability that turns official information into verified, multi-channel public communication within minutes.